# Part A: Conceptual Foundation (Theory)

**1. What are Ensemble Learning Methods and why are they effective?**

Ensemble Learning is a machine learning technique that combines multiple models to make one final prediction. Instead of relying on a single model, it uses the strengths of several models. This improves prediction accuracy, reduces errors, and makes the model more reliable.

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**2. Explain the difference between Bagging, Boosting, and Voting.**

- **Bagging** trains multiple models independently on different random subsets of the data and combines their predictions.
- **Boosting** trains models one after another, where each new model focuses on correcting the mistakes of the previous model.
- **Voting** combines predictions from different trained models and selects the final prediction using majority voting (classification) or averaging (regression).

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**3. What is Bias–Variance Trade-off, and how do ensemble methods address it?**

Bias is the error caused by overly simple models, while variance is the error caused by models being too sensitive to training data. A good model balances both. Bagging mainly reduces variance, while boosting reduces bias, resulting in better overall performance.

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**4. Explain Voting Classifier and Stacking Ensemble.**

A **Voting Classifier** combines predictions from multiple classifiers using hard voting (majority vote) or soft voting (highest average probability). A **Stacking Ensemble** combines several base models and uses another model (meta-learner) to learn how to combine their predictions for better accuracy.

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**5. Compare AdaBoost, Gradient Boosting, LightGBM, and XGBoost conceptually.**

| Algorithm | Description |
|---|---|
| **AdaBoost** | Gives more importance to incorrectly classified samples and trains weak learners sequentially. |

| Algorithm | Description |
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| **Gradient Boosting** | Builds each new model to reduce the prediction errors of previous models using gradient descent. |
| **LightGBM** | A fast and memory-efficient gradient boosting algorithm designed for large datasets. |
| **XGBoost** | An optimized gradient boosting algorithm that provides high accuracy, regularization, and faster training. |